

EEG-Based Motor Imagery Classification Using Machine Learning Techniques for Brain-Computer Interface Applications



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Brain-Computer Interfaces (BCIs) have become a promising technology for restoring and motor function in individuals with neurological disorders. Among the different BCI paradigms, Motor Imagery (MI) has attracted considerable attention due to its potential application in neurorehabilitation and assistive technologies. This study presents the development and evaluation of a machine learning pipeline for the classification of electroencephalographic (EEG) signals associated with three motor imagery conditions: left-hand imagery, right-hand imagery, and resting state.

The proposed framework includes signal preprocessing, artifact rejection, feature extraction, feature selection, and supervised classification. Two ERD+MRCP feature models and a Common Spatial Pattern (CSP) cascade architecture were evaluated using within-subject and Leave-One-Subject-Out (LOSO) cross-validation strategies. Feature analysis by implementing Kruskal-Wallis tests and Cohen's d effect sizes revealed that MRCP features when looking at Fz and Cz, are most discriminative for differentiating rest or movement, while Mu-band ERD lateralization between C3 and C4 is more definitive for right versus left discrimination, motivating a two-stage cascade classification approach.

The CSP cascade achieved the best performance, reaching an accuracy of $59.0 \pm 8.9\%$ within subject and $49.4 \pm 8.9\%$ with LOSO, compared to the 33.3% chance at a random selection. The primary limitation was the low recall of the rest class (0.36), driven by the weak effect sizes of features available to the cascade's first stage. Inter-subject variability was high, with individual LOSO accuracies ranging from near chance to 65%. Although the latency analysis confirmed that the pipeline is computationally negligible (< 1.3 ms), the 1500 ms analysis window renders it unsatisfactory for real-time BCI implementation. Further adaptation is needed.

These findings contribute to the development of robust EEG-based decoding systems and support the potential use of Motor Imagery BCIs in future neurorehabilitation and assistive technology scenarios.

1 Introduction

Brain-Computer Interfaces (BCIs) are systems that establish a direct communication pathway between the human brain and external devices by translating neural activity into control commands. Over the last decades, BCIs have gained considerable attention due to their potential applications in assistive technologies, neuroprosthetics, and neurorehabilitation. By bypassing conven-

tional neuromuscular pathways, these systems can provide communication and motor control capabilities to individuals affected by neurological disorders such as stroke, spinal cord injury, amyotrophic lateral sclerosis, or traumatic brain injury.

Among the various BCI paradigms, Motor Imagery (MI) has emerged as one of the most widely investigated approaches. Motor imagery refers to the mental simulation of a movement without any actual muscu-

lar execution. Neurophysiological studies have demonstrated that imagined and executed movements share similar cortical activation patterns, particularly within sensorimotor regions. This characteristic makes MI especially attractive for rehabilitation applications, as repeated mental practice can promote neuroplasticity and facilitate motor recovery in patients with motor impairments.

Electroencephalography (EEG) is one of the most commonly used signal acquisition techniques in BCI research due to its non-invasive nature, low cost, portability, and high temporal resolution. During motor imagery tasks, specific changes in brain rhythms can be observed over sensorimotor areas. In particular, Event-Related Desynchronization (ERD) and Event-Related Synchronization (ERS) phenomena are typically detected within the Mu (8–13 Hz) and Beta (14–30 Hz) frequency bands. These modulations provide valuable information for distinguishing between different imagined movements and constitute the basis of many EEG-based BCI systems.

Despite significant advances in the field, accurate decoding of motor intentions from EEG recordings remains a challenging task. EEG signals are characterized by a low signal-to-noise ratio, high inter-subject variability, and susceptibility to artifacts originating from eye movements, muscle activity, and environmental noise. Consequently, the performance of a BCI system strongly depends on the effectiveness of signal preprocessing, feature extraction, feature selection, and classification methodologies.

The development of robust machine learning techniques has significantly improved the performance of EEG-based classification systems. Traditional approaches such as Linear Discriminant Analysis (LDA), Support Vector Machines (SVM), and Random Forests have demonstrated promising results when combined with appropriate feature extraction techniques, including spectral analysis, Common Spatial Patterns (CSP). However, selecting the most suitable combination of features and classifiers remains an active area of research.

The objective of this study is to develop and evaluate a complete machine learning pipeline for the classification of motor imagery EEG signals. Specifically, three classes are considered: left-hand motor imagery, right-hand motor imagery, and resting state. The proposed framework includes signal preprocessing, exploratory data analysis, feature extraction, feature selection, model training, and performance evaluation. Furthermore, the computational requirements of the devel-

oped pipeline are analysed to assess its potential suitability for real-time Brain-Computer Interface applications.

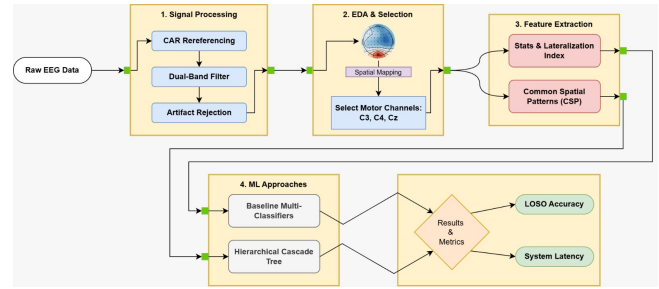


Figure 1: Overview of the proposed motor imagery EEG classification framework. Raw EEG signals undergo preprocessing, exploratory data analysis and channel selection, followed by two alternative feature extraction approaches: handcrafted ERD/MRCP features and Common Spatial Patterns (CSP). The extracted features are evaluated using baseline machine learning classifiers and a hierarchical cascade architecture. Final performance is assessed using LOPO accuracy and computational latency metrics.

The remainder of this paper is organized as follows. Section II describes the dataset, experimental protocol, and methodology employed throughout the study. Section III presents the obtained results. Section IV discusses the findings, limitations, and potential implications of the proposed approach. Finally, Section V summarizes the main conclusions and outlines future research directions.

2 Materials and Methods

This section describes the dataset, signal preprocessing procedures, feature extraction techniques, classification models, and evaluation methods employed throughout the study. The proposed workflow was designed to transform raw electroencephalographic (EEG) recordings into meaningful representations suitable for motor imagery classification.

2.1 Dataset Description

The dataset used in this study consisted of EEG recordings acquired from healthy volunteers performing motor imagery tasks. Data were collected using a 16-channel OpenBCI EEG acquisition system with a sampling frequency of 125 Hz. The EEG signals were referenced to the right mastoid during acquisition.

Although the recording system provided 16 channels, only the first 15 electrodes were considered for analy-

sis, while the sixteenth channel was discarded as recommended by the dataset documentation. The electrode locations followed the international 10–20 system and included the following channels: F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, P7, P3, Pz, P4, and P8.

The experimental protocol consisted of eight runs of sixty trials each, resulting in a total of 480 trials per subject. The first two runs corresponded to motor execution tasks, whereas the remaining six runs contained motor imagery tasks. Since the objective of this study was the classification of motor imagery states, only the motor imagery trials were considered during the analysis.

Consequently, a total of 360 motor imagery trials were available for each participant. These trials were equally distributed among three classes:

- Left-hand motor imagery (120 trials)
- Right-hand motor imagery (120 trials)
- Rest condition (120 trials)

2.2 Experimental Protocol

The experimental paradigm was implemented using PsychoPy and followed a cue-based motor imagery design. Each trial lasted approximately five seconds and was composed of three consecutive stages.

First, a fixation cross was displayed to the participant for approximately five seconds, with a small random temporal variation added to reduce anticipation effects. Following this period, a visual cue instructed the participant to perform either left-hand motor imagery, right-hand motor imagery, or remain at rest. The task period lasted two seconds. Finally, an inter-trial interval of approximately two seconds was introduced before the beginning of the next trial.

The order of the tasks was randomized within each run to minimize potential ordering effects and maintain participant attention throughout the experiment.

2.3 Signal Preprocessing

EEG preprocessing was performed to improve signal quality and enhance the discriminative neural activity associated with motor imagery.

Initially, all signals were detrended to remove constant offsets and low-frequency drifts. Subsequently, four band-pass filtered versions of the EEG signals were generated. An 8–30 Hz filter was used for general ERD analysis, an 8–13 Hz filter for Mu-band activity, a 14–30 Hz filter for Beta-band activity, and a 0.3–3 Hz filter

for Movement-Related Cortical Potential (MRCP) analysis.

Bad channels were automatically identified using statistical criteria and excluded from subsequent spatial filtering procedures. A Common Average Reference (CAR) was then applied to reduce spatially correlated noise and improve the signal-to-noise ratio across electrodes.

The filtered EEG recordings were then segmented into epochs corresponding to the motor imagery interval. Each epoch consisted of a two-second window starting at the onset of the visual cue. For ERD analysis, only the central interval between 0.5 s and 1.5 s after cue onset was retained, as this period contains the most stable motor imagery activity. For MRCP analysis, a one-second window centered around movement onset (-0.5 s to $+0.5$ s) was extracted. Artifact-contaminated trials were automatically rejected using adaptive subject-specific thresholds. For each participant, the 95th percentile of the maximum amplitude and standard deviation distributions was computed from channels C3, Cz, and C4. A trial was discarded when at least two of these channels exceeded either threshold.

2.4 Exploratory Data Analysis

An exploratory analysis was performed to assess signal quality and verify the presence of motor imagery-related neural activity.

Initially, raw EEG signals were visually compared with their filtered counterparts in both the ERD (8–30 Hz) and MRCP (0.3–3 Hz) frequency ranges. Subsequently, subject-specific trial rejection thresholds were inspected through amplitude and standard deviation distributions.

In addition, grand-average topographic maps were generated across all subjects for Mu-band ERD, Beta-band ERD, and MRCP activity. These spatial representations were used to verify the expected sensorimotor activation patterns associated with left-hand motor imagery, right-hand motor imagery, and rest conditions.

2.5 Feature Engineering

Feature extraction was performed to transform the pre-processed EEG signals into representative variables suitable for machine learning classification.

Feature extraction focused on two complementary neurophysiological phenomena commonly used in motor imagery Brain-Computer Interfaces: Event-Related Desynchronization (ERD) and Movement-Related Cortical Potentials (MRCPs).

ERD features were computed from relative power variations in the Mu (8–13 Hz) and Beta (14–30 Hz) frequency bands with respect to a baseline period. MRCP features were extracted from low-frequency EEG activity (0.3–3 Hz) using temporal descriptors that characterize movement preparation and execution processes. Furthermore, Common Spatial Patterns (CSP) were employed to maximize the variance differences between classes and enhance class separability.

Two baseline feature extraction approaches and several CSP-based approaches were investigated in this study.

2.5.1 Model 1: ERD and Single-Electrode MRCP Features

The first model combined Event-Related Desynchronization (ERD) features with Movement-Related Cortical Potential (MRCP) features.

ERD features were extracted from the sensorimotor electrodes C3, Cz, and C4. For each electrode, ERD values were computed within the Mu (8–13 Hz) and Beta (14–30 Hz) frequency bands relative to a baseline period. Additionally, lateralization indices were calculated for both frequency bands in order to capture hemispheric differences associated with left- and right-hand motor imagery.

The extracted ERD features included:

- Mu-band ERD at C3, Cz, and C4
- Beta-band ERD at C3, Cz, and C4
- Mu-band lateralization index
- Beta-band lateralization index

MRCP features were extracted from electrodes Cz and Fz, as this location is strongly associated with movement preparation and motor planning processes. The following descriptors were computed:

- Minimum amplitude
- Maximum amplitude
- Area under the curve (AUC)
- Standard deviation

This model was designed to combine frequency-domain information derived from ERD measurements with temporal characteristics extracted from MRCP signals while relying on two central electrodes (Cz and Fz) for movement-related potential analysis.

2.5.2 Model 2: ERD and Multi-Electrode MRCP Features

The second model employed the same ERD features as Model 1 but expanded the MRCP analysis to include spatial information from four sensorimotor electrodes: C3, Cz, Fz and C4.

The same MRCP descriptors (minimum amplitude, maximum amplitude, area under the curve, and standard deviation) were extracted from each electrode individually. This model was designed to evaluate whether additional spatial information could improve classification performance compared with the single-electrode approach.

2.5.3 CSP-Based Models

In addition to the two baseline models, several Common Spatial Pattern (CSP) configurations were evaluated.

Different combinations of frequency bands and channel configurations were investigated, including:

- Mu band (8–13 Hz)
- Beta band (14–30 Hz)
- Combined Mu and Beta bands
- Full ERD frequency range

Furthermore, CSP filters were computed using both three-channel and eighth-channel electrode configurations to evaluate the effect of spatial information on classification performance.

2.5.4 CSP Cascade Architecture

Since CSP is inherently a binary classification technique, a hierarchical cascade architecture was implemented to address the three-class classification problem.

The first stage discriminated between Rest and Movement conditions. Subsequently, trials classified as Movement were processed by a second CSP classifier responsible for discriminating between Left-Hand and Right-Hand motor imagery.

The overall decision pipeline can be summarized as follows:

Rest vs Movement → Left vs Right

This cascade approach allowed CSP to be effectively applied to a multiclass motor imagery problem while preserving its discriminative capabilities.

2.6 Topographic Analysis

To investigate the spatial distribution of motor imagery-related activity, grand-average topographic maps were generated across all subjects.

Separate scalp maps were computed for Mu-band ERD (8–13 Hz), Beta-band ERD (14–30 Hz), and MRCP activity. For each signal type, topographies were generated for left-hand motor imagery, right-hand motor imagery, and rest conditions. Additionally, difference maps were computed to highlight the spatial patterns associated with motor imagery lateralization.

2.7 Feature Selection and Dimensionality Reduction

Different dimensionality reduction strategies were applied depending on the classifier and feature set.

For the baseline ERD+MRCP models, Principal Component Analysis (PCA) was applied when using Linear Discriminant Analysis (LDA). This decision was motivated by the presence of redundant and highly correlated features, which can negatively affect LDA performance.

Similarly, for CSP-based models, feature selection procedures were applied prior to LDA classification in order to retain only the most informative CSP components and reduce redundancy.

In contrast, Support Vector Machines (SVM), Logistic Regression (LR), and Random Forests (RF) were trained directly on the extracted feature sets without additional dimensionality reduction. These algorithms are generally more robust to correlated and redundant features and therefore do not require explicit feature reduction strategies.

2.8 Classification Models

Four supervised machine learning algorithms were evaluated throughout the study:

- Linear Discriminant Analysis (LDA)
- Support Vector Machine (SVM)
- Logistic Regression (LR)
- Random Forest (RF)

These classifiers were selected due to their extensive use in Brain-Computer Interface research and their ability to handle EEG-based classification problems with different feature representations.

2.9 Validation Strategy and Performance Metrics

Two complementary evaluation strategies were implemented to assess classifier performance.

2.9.1 Within-Subject Evaluation

The first evaluation was performed independently for each subject using a stratified train-test split. Approximately 80% of the trials were used for training, while the remaining 20% were reserved for testing.

This approach allows the classifier to learn subject-specific neural patterns and typically produces higher classification performance. However, it does not provide information regarding generalization across different individuals.

2.9.2 Leave-One-Subject-Out Evaluation

To assess inter-subject generalization, a Leave-One-Subject-Out (LOSO) cross-validation scheme was implemented.

Given the dataset consisted of 11 subjects, 11 folds were generated. In each fold, data from 10 subjects were used for training, while the remaining subject was used exclusively for testing. This process was repeated until every participant had served as the test subject exactly once.

Compared with within-subject evaluation, LOSO provides a more realistic estimate of the performance expected in practical Brain-Computer Interface applications, where the system must generalize to previously unseen users.

Classification performance was assessed using accuracy, precision, recall, F1-score, and confusion matrices. In addition, computational performance was analysed to evaluate the suitability of the proposed pipeline for future real-time BCI applications.

3 Results

3.1 Exploratory Analysis Results

Before feature extraction, signal quality was assessed using a subject-adaptive artifact rejection strategy based on two criteria computed across three electrodes that are connected to motor sensing: C3, Cz, and C4. The two aforementioned criteria are the 95th-percentile peak amplitude and the 95th-percentile standard deviation of the bandpass-filtered signal within each trial window. The thresholds varied across participants, reflecting the well-known variability between subjects in the EEG

amplitude, ranging from 3.10 μV to 8.45 μV for peak amplitude and from 0.93 μV to 2.44 μV for standard deviation. Figure 2 illustrates the trial-by-trial distribution of both metrics for an example subject (v01), along with the corresponding rejection thresholds.

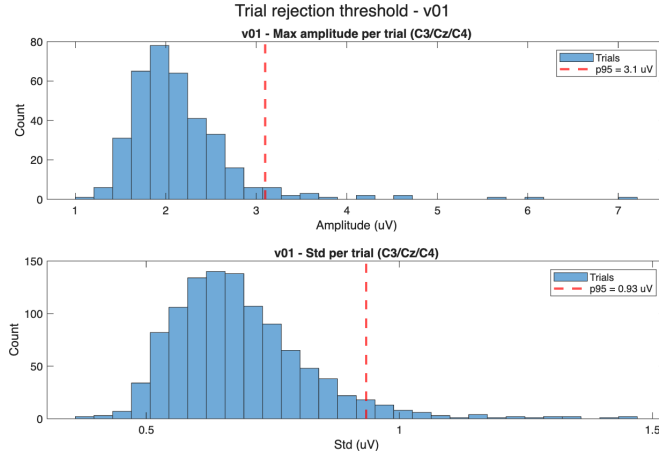


Figure 2: Trial rejection thresholds for subject v01. Top: distribution of per-trial maximum amplitude across electrodes C3, Cz, and C4, filtered in the 8-30 Hz band; the dashed red line marks the 95th-percentile threshold (3.10 μV). Bottom: corresponding distribution of per-trial standard deviation (threshold: 0.93 μV). Trials exceeding either threshold were excluded.

Across all 11 participants, the mean rejection rate was $4.2 \pm 0.9\%$, indicating that the majority of trials were retained for analysis (Table 1). The balance of the class was broadly preserved after rejection, with the number of trials accepted per condition remaining close to the original 120.

Table 1: Trial rejection summary per subject. Thresholds were set at the 95th percentile of peak amplitude and standard deviation across the sensorimotor electrodes.

Subject	Accepted	Rejected (%)	Right	Left	Rest
v01	341	5.3	118	116	107
v02	346	3.9	116	111	119
v03	345	4.7	118	116	111
v04	343	4.7	115	112	116
v05	348	3.3	115	117	116
v06	350	2.8	120	115	115
v07	350	3.0	120	119	111
v08	342	5.3	116	110	116
v09	349	3.3	120	116	113
v10	343	5.0	117	117	109
v11	343	5.0	118	118	107
Mean	345	4.2 ± 0.9	-	-	-

Figure 3 shows the accepted and rejected trial counts

per subject in stacked bar form. The low and consistent rejection rate across participants confirms that recording quality is satisfactory and that the experimental protocol has successfully been followed.

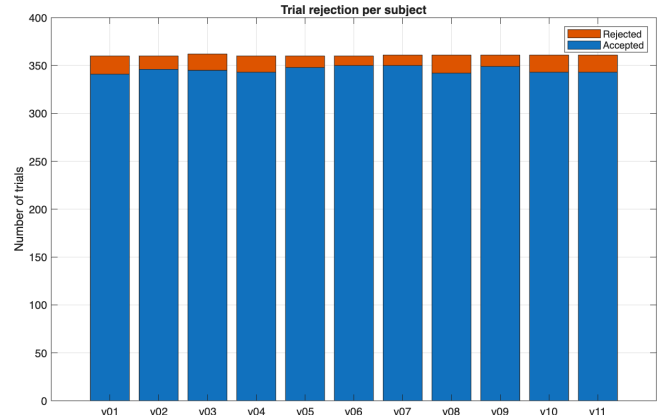


Figure 3: Stacked bar chart of accepted (blue) and rejected (orange) trials per subject. On average, $4.2 \pm 0.9\%$ of trials were excluded, preserving the intended class balance.

The two filtering operations applied in this pipeline target fundamentally different neural phenomena from the same raw signal, and their effect is visualised in Figures 4 and 5 for an example trial. The 8-30 Hz band-pass filter (Figure 4) isolates the rhythmic Mu and Beta oscillations associated with ERD/ERS without distorting the underlying signal: the oscillatory envelope is cleanly extracted while the slow drift visible in the raw trace is suppressed. The 0.3-3 Hz filter (Figure 5) acts on the same raw signal but captures an entirely different component: the slow cortical potential waveform associated with motor preparation, exhibiting the characteristic negative deflection followed by a positive peak centred near 0.8-1.0 s post-cue, consistent with MRCP morphology reported for motor imagery tasks [1]. Crucially, both filters are applied to the same raw recording without any signal loss between them, confirming that the ERD and MRCP feature families are extracted from complementary frequency content of the same EEG source.

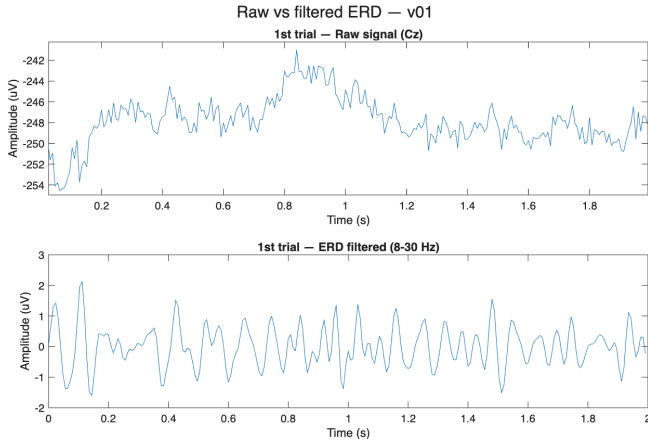


Figure 4: Raw EEG (top) and 8-30 Hz bandpass-filtered signal (bottom) at electrode Cz for the first motor imagery trial of subject v01. Filtering isolates the Mu and Beta band oscillatory content without distorting the signal, validating its use for ERD/ERS feature extraction.

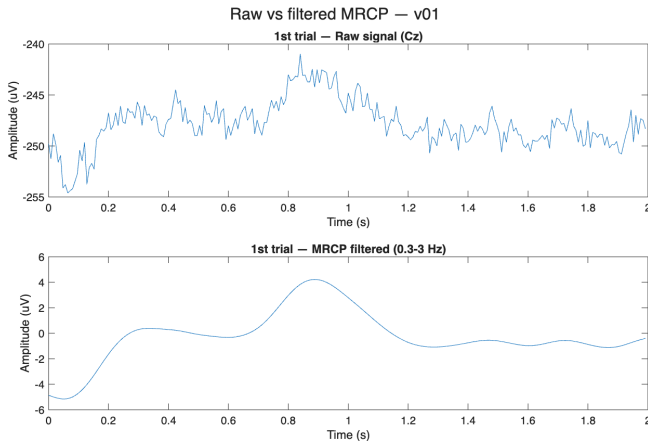


Figure 5: Raw EEG (top) and 0.3-3 Hz filtered signal (bottom) at electrode Cz for the same trial as Figure 4. The slow cortical waveform reveals a distinct biphasic shape not visible in the raw signal, demonstrating that the two filters extract different physiological phenomena from the same recording.

The spatial distribution of the filtered signals across the scalp is shown in the topographic maps of Figures 6, 7, and 8, computed as grand averages across all subjects. The Mu and Beta ERD maps show bilateral desynchronization over the central sensorimotor electrodes C3 and C4, with a clear hemispheric asymmetry in the left-minus-right difference map, consistent with the expected contralateral pattern. The MRCP topographic maps show the slow cortical activity to be maximal at frontal and central midline locations, particularly Fz and Cz, with no meaningful lateralization between left and right imagery. The movement-minus-rest contrast confirms that the MRCP response is centrally distributed and not hemispheric, further motivating the inclusion of

Fz as an additional MRCP electrode alongside Cz.

The left minus right difference maps confirm that the lateralization effect is spatially specific. In the Mu band the asymmetry peaks at C3 and C4 with a difference of approximately 0.3 normalized units. Additionally, in the Beta band the pattern is weaker but consistent, providing enough justification for the use of both bands as complementary lateralization features.

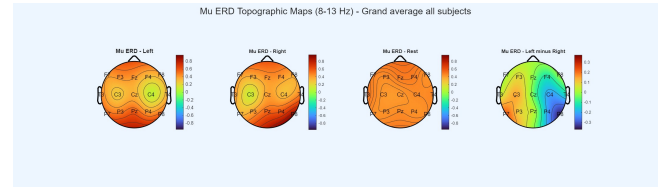


Figure 6: Grand-average Mu-band ERD topographic maps (8-13 Hz) for left imagery, right imagery, rest, and the left-minus-right difference, averaged across all 11 subjects. The difference map confirms the expected contralateral de-synchronization pattern over C3 and C4.

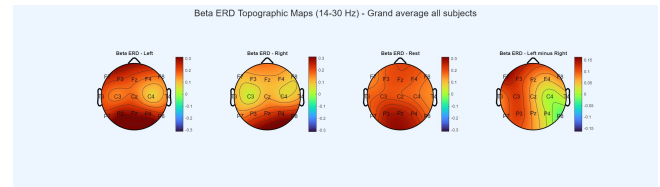


Figure 7: Grand-average Beta-band ERD topographic maps (14-30 Hz) for left imagery, right imagery, rest, and the left-minus-right difference. The lateralized pattern mirrors the Mu band, supporting the use of both bands for lateralization-based discrimination.

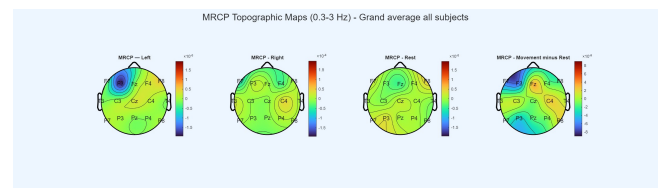


Figure 8: Grand-average MRCP topographic maps (0.3-3 Hz) for left imagery, right imagery, rest, and the movement-minus-rest difference. The slow cortical activity is maximal at frontal and central midline locations (Fz, Cz), with no hemispheric asymmetry, motivating the inclusion of Fz as an additional MRCP feature electrode.

3.2 Feature Analysis Results

The discriminative power of each feature was assessed using the Kruskal-Wallis non-parametric test and Cohen's d effect size, evaluated separately for the two bi-

nary sub-problems that underpin the cascade classification strategy: rest vs. movement, and left vs. right hand imagery. Figure 9 presents both metrics simultaneously, making it possible to identify which features are statistically significant, which carry a meaningful practical effect, and how the two sub-problems call for different feature families.

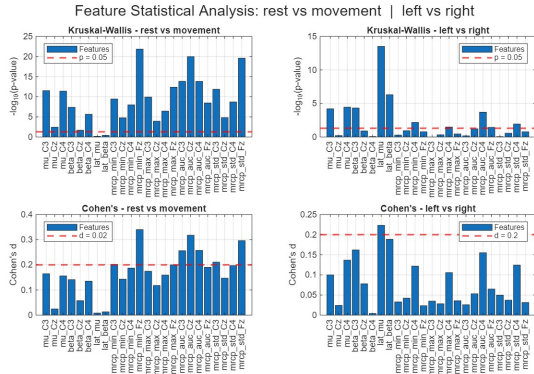


Figure 9: Feature discriminability analysis for the rest vs. movement (left column) and left vs. right (right column) sub-problems. Top row: $-\log_{10}(p)$ values from Kruskal-Wallis tests; the dashed red line marks $p = 0.05$. Bottom row: corresponding Cohen's d effect sizes; the dashed red line marks $d = 0.2$. ERD features dominate the left-vs.-right problem while MRCP features contribute primarily to rest-vs.-movement detection, motivating the cascade architecture.

The MRCP features at Cz and Fz reach the highest statistical significance, with the minimum amplitude at Fz being among the most discriminative slow-cortical-potential descriptors, alongside Mu-band ERD at C3 and C4. This confirms the significance of slow cortical activity at the central and frontal midline electrodes as a meaningful signal for detecting motor imagery onset. Such a result is consistent with the role of movement-related cortical potentials in motor preparation [1]. Figure 10 below makes the separation directly visible. There is a systematic displacement in the rest condition when compared to both of the movement classes. On the other hand, the left and right movements are indistinguishable based on this feature.

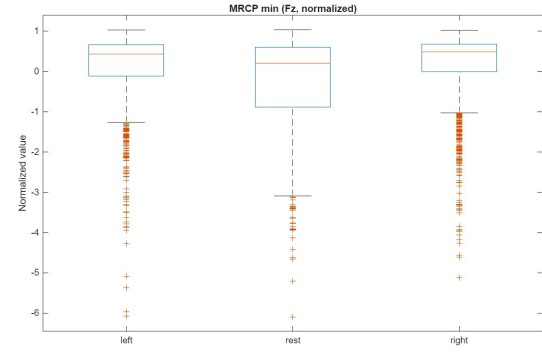


Figure 10: Boxplot of the normalised MRCP minimum amplitude at Fz for the three classes, pooled across all subjects. Rest is displaced from both movement conditions, consistent with its significant Kruskal-Wallis statistic for the rest vs. movement sub-problem. Left and right are virtually identical, confirming this feature carries no lateralization information.

Moving on to the left vs. right discrimination, the Mu-band lateralization index, or C3 – C4, is the dominant feature: $d \approx 0.23$, $-\log_{10}(p) \approx 13.5$. The next feature is the Beta lateralization index - $d \approx 0.19$. MRCP features are largely non-significant for this sub-problem, confirming that hemispheric asymmetry in the sensorimotor rhythm is the primary source of left/right discriminability [2], rather than slow cortical potentials. Figure 11 shows the distribution of the Mu lateralization index restricted to the two movement classes. Filtering out the rest condition makes the directional difference between left and right more apparent.

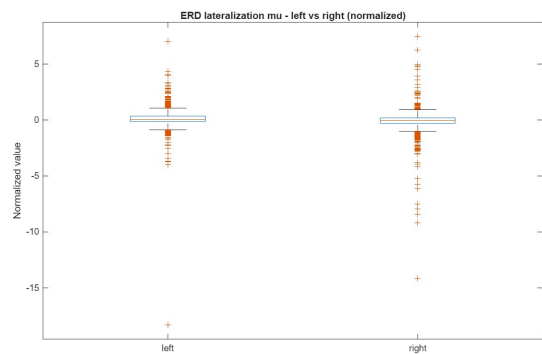


Figure 11: Boxplot of the normalised Mu-band ERD lateralization index (C3 – C4) for left and right imagery classes only, pooled across all subjects. The directional difference between conditions is consistent with the expected contralateral de-synchronization pattern, corresponding to the $d \approx 0.23$ effect size identified in Figure 9. Considerable overlap remains, reflecting the single-trial classification difficulty.

Together, these two figures justify the complementary roles of ERD and MRCP features and provide motivation for the cascade architecture. Firstly, using MRCP-dominant features to separate movement from rest state. Secondly, ERD lateralization is used to discriminate between left and right.

The complementary nature of these features is further confirmed by the correlation matrix in Figure 12. On the matrix two distinct blocks of high correlation are visible. one for the ERD features and one for the MRCP descriptors, with negligible cross-block correlation ($r \approx 0$). The ERD block exhibits moderate-to-high correlations among electrodes and bands, motivating the use of PCA before LDA to reduce redundancy. While, the MRCP block shows that intra-electrode descriptors (min, max, AUC, std) are highly correlated, while between-electrode correlations are lower, suggesting that the spatial expansion from one electrode (Model 1) to four (Model 2) introduces genuinely new information.

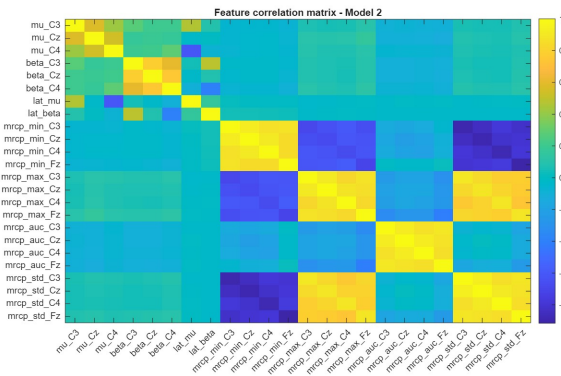


Figure 12: Pearson correlation matrix of the 24 features in Model 2, pooled across all subjects. Two main blocks are visible: ERD features (top-left, 8 features) and MRCP descriptors (bottom-right, 16 features including Fz). Near-zero cross-block correlations confirm that the two families provide complementary, non-redundant information, while high intra-block correlations within the ERD group motivate dimensionality reduction prior to LDA.

Further a scatter plot is made to of the two most discriminative features from each sub-problem. Here Mu lateralization index, best for left vs. right, is plotted against the MRCP minimum amplitude at Fz, best MRCP feature for rest vs. movement. The result is shown in Figure 13. It is visible that the classes overlap significantly, however the two axes vary independently. This characteristic of the data provides proof while the combination is beneficial, meaning that one type provides information that the other does not.

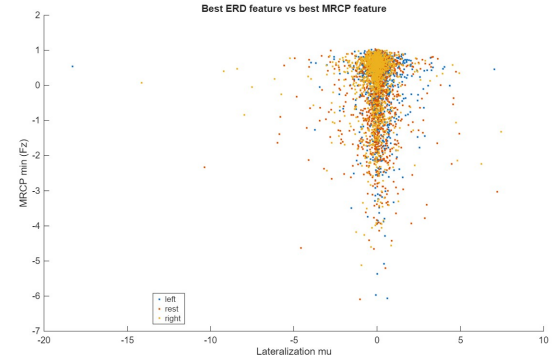


Figure 13: Scatter plot of the Mu-band lateralization index (C3 – C4) versus the MRCP minimum amplitude at Fz, coloured by class. The horizontal axis captures the best feature for left-vs.-right; the vertical axis the best MRCP feature for rest-vs.-movement. The two axes vary independently, consistent with the near-zero cross-block correlation in Figure 12.

Finally, the PCA projection of the full Model 2 feature space onto its first two principal components in Figure 14 shows that, while no sharp cluster boundaries are visible in the 2D view, the classes are not randomly interleaved. This indicates that discriminative information exists in the data but is distributed across multiple dimensions rather than concentrated in a single axis, providing a reason for using all features jointly in the classifier rather than selecting a small subset.

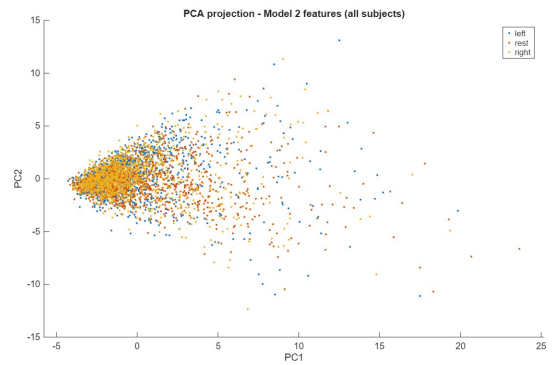


Figure 14: PCA projection of the Model 2 feature set onto the first two principal components, pooled across all subjects. The absence of tight clusters confirms that the problem is not linearly separable in a low-dimensional projection, but the gradual class structure along PC1 indicates that discriminative information is present and distributed across the feature space, justifying the use of the full feature vector.

3.3 Classification Results

The following section will discuss the results of the classification and the strategies used to obtain them. The classification performance is evaluated using within-subject evaluation, stratified 80/20 hold-out per subject, and Leave-One-Subject-Out (LOSO) cross-validation. The three-class chance level is equal to 33.3%. Table 2 supplies the values of the best configuration per pipeline under each strategy, while figure 15 provides a visual comparison.

Table 2: Best mean accuracy ($\pm$ std, 11 subjects) per pipeline. Chance level: 33.3%.

Pipeline	Best within model	Within acc.	Best LOSO model	LOSO acc.
Model 1 (ERD+MRCP Cz)	M1_LDA	$46.6 \pm 8.4\%$	M1_LogReg	$42.6 \pm 7.4\%$
Model 2 (ERD+MRCP C3/Cz/C4)	M2_LogReg	$48.4 \pm 9.9\%$	M2_LDA	$44.5 \pm 8.5\%$
CSP cascade	8ch base cascade	$59.0 \pm 8.9\%$	8ch mubeta cascade	$49.4 \pm 8.9\%$



Figure 15: Mean within-subject (blue) and LOSO (orange) accuracy for the best model of each pipeline. The dashed line marks the 33.3% chance level. The gap between the two bars for each pipeline quantifies the loss of generalisation to unseen subjects.

All pipelines reach a score well over the chance value of 33.3%. The ERD+MRCP models achieved 46.6–48.4% within-subject, dropping to 42.6–44.5% under LOSO. The difference between Model 1 and Model 2 is small but consistent (~ 2 pp), confirming that adding spatial MRCP information at C3 and C4 provides a modest benefit.

The CSP cascade classifier outperforms both ERD+MRCP approaches under every condition, reaching 59.0% within-subject and 49.4% under LOSO. This represents a gain of approximately 10 pp over the ERD+MRCP baseline in the within-subject setting, and roughly 5 pp under LOSO. The larger gap between within-subject and LOSO performance observed in the CSP cascade (~ 9 pp) compared to Approaches 1 and 2 (~ 4 pp) suggests that the spatial filters learned by CSP are more subject-specific and generalise less well to unseen individuals. Nonetheless, all three approaches consistently surpass chance under both validation strategies, confirming the validity of the

proposed pipeline.

Figure 16 shows the within-subject and LOSO accuracy distributions side by side across all ERD+MRCP classifiers. Across the classifiers the performance is overall similar. The Random Forest is consistently lowest and LDA/LogReg at the top. This finding suggests that the feature space is better suited to linear boundaries. The large within-pipeline variance ($\sigma \approx 8$ –10%) reflects high inter-subject variability. The LOSO results suggest that this characteristic is not captured by any model.

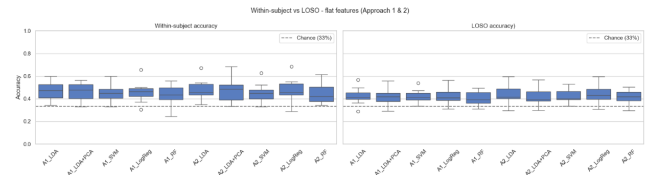


Figure 16: Within-subject (left) and LOSO (right) accuracy distributions for all ERD+MRCP classifiers. The dashed line marks chance. All models exceed chance under both strategies, but variability is large, and no classifier stands out clearly from the rest.

The CSP cascade outperformed both baseline models presenting results of $59.0 \pm 8.9\%$ within-subject and $49.4 \pm 8.9\%$ under LOSO. The 8-channel configurations consistently exceed their 3-channel counterparts, indicating that electrodes outside the central motor strip contribute useful spatial information. Figure 18 shows the within-subject and LOSO accuracy for all CSP cascade configurations. The combined Mu+Beta band configurations perform best in both evaluation strategies. For reference, CSP with direct three-class LDA (rather than the cascade) reached $72.0 \pm 8.5\%$ within subject, substantially higher than any cascade result. The cascade introduces first stage errors that propagate to the second stage and cap overall performance.

Such a gap is explained by the two stage nature of the architecture. If the problem is segmented into two separate tasks, differentiating the rest and movement stages and then out of the movement labels, left or right is decided, whatever error is made in the first estimation gets carried over to the second one. A direct multi-class CSP model avoids this by optimising all three boundaries simultaneously, which is why it achieves substantially higher accuracy despite using the same spatial filters.

Most frequent classifier per step - Within-subject (8ch_base_cascade):
 CSP1_rest_vs_move: 8ch_base_CSP+LDA (selected in 7/11 folds, mean acc = 0.745)
 CSP2_left_vs_right: 8ch_base_CSP+LogReg (selected in 5/11 folds, mean acc = 0.707)

Most frequent classifier per step - LOSO (8ch_mubeta_cascade):
 CSP1_rest_vs_move: 8ch_mubeta_CSP+LDA+FS (selected in 6/11 folds, mean acc = 0.700)
 CSP2_left_vs_right: 8ch_mubeta_CSP+LDA (selected in 5/11 folds, mean acc = 0.620)

Figure 17: Most frequently selected classifier configuration at each stage of the CSP cascade. The first stage (rest vs. movement) consistently favours LDA-based models, whereas the second stage (left vs. right) benefits from either Logistic Regression or LDA depending on the evaluation strategy. Results are reported for both within-subject and LOSO analyses.

To better understand the behaviour of the cascade architecture, the most frequently selected classifier at each stage was analysed across all subjects and LOSO folds (Figure 17). For within-subject evaluation, the rest-versus-movement stage was most often solved using an 8-channel CSP representation combined with LDA, achieving a mean accuracy of 74.5%. In contrast, the left-versus-right stage favoured Logistic Regression, reaching a mean accuracy of 70.7%.

Under LOSO evaluation, the optimal configuration shifted towards Mu+Beta CSP features. The rest-versus-movement classifier remained LDA-based and achieved a mean accuracy of 70.0%, while the left-versus-right stage also converged towards LDA with a mean accuracy of 62.0%. These results suggest that the two stages of the cascade benefit from different classifier-feature combinations and justify the use of stage-specific optimisation rather than a single classifier throughout the entire pipeline.

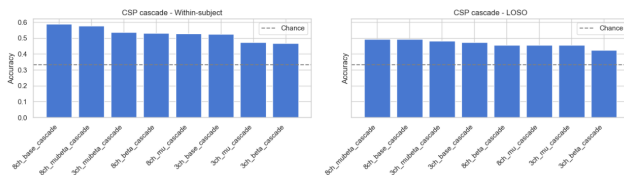


Figure 18: Within-subject (left) and LOSO (right) accuracy for all CSP cascade configurations. The 8-channel configurations consistently outperform their 3-channel counterparts, and combined Mu+Beta band configurations perform best under both evaluation strategies.

The LOSO accuracy per subject for the best overall model, or 8ch mubeta cascade, is shown in Figure 19. The performances variate from near chance for subjects v01 and v04/v05 to above 65% for subject v06. These findings illustrate the inter-subject variability that is the suspected limitation of this approach.

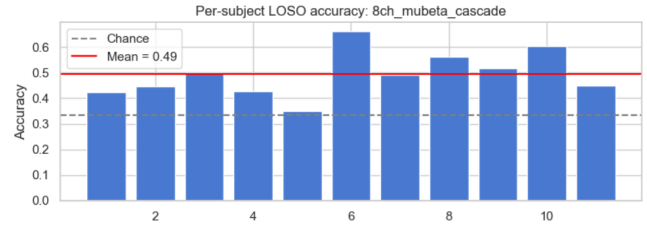


Figure 19: Per-subject LOSO accuracy for the 8ch mubeta cascade model. The red line marks the cross-subject mean (0.49); the dashed line marks chance. Three subjects fall near or at chance, while others reach 57-65%, demonstrating the high inter-subject variability that is characteristic of motor imagery EEG.

Figures 20 and 21 present confusion matrix and per-class metrics for the best LOSO model. They reveal the structure of the errors. The model confuses left and right imagery symmetrically, while rest is the hardest class to detect: recall drops to 0.36, meaning roughly 64% of rest trials are misclassified as a movement class. This asymmetry is consistent with the feature analysis. Overall, a leak of rest samples into the movement branch is suggested by the fact that the cascade's first stage relies on ERD and MRCP features with effect sizes of $d \leq 0.31$, equalling the weakest in the entire feature set.

It is noteworthy to mention that within the misclassifications 433 trials are predicted as left compared to 357 as right. Such an asymmetric characteristic suggests a mild left side bias, making population-level tendency toward left-lateralized resting-state activity in the sensorimotor cortex a viable possibility. Findings of such nature warrant further investigation on a larger cohort.

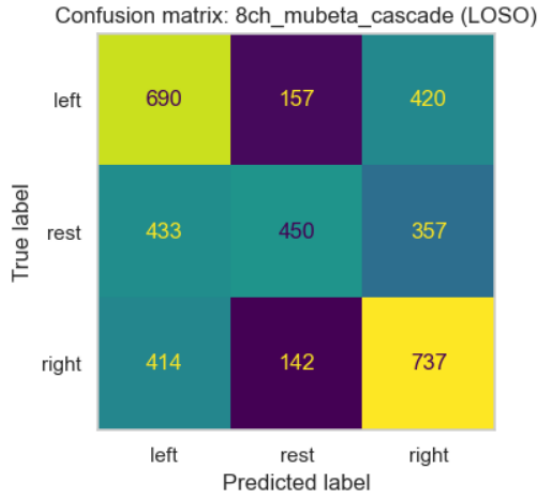


Figure 20: Aggregate confusion matrix for the 8ch mubeta cascade under LOSO evaluation (counts pooled across all 11 folds). Left and right imagery are classified with reasonable symmetry, while rest is largely misclassified as a movement condition.

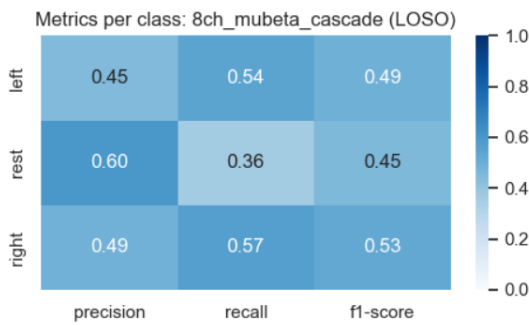


Figure 21: Per-class precision, recall, and F1-score for the 8ch mubeta cascade (LOSO). Left and right achieve F1 scores of 0.49 and 0.53 respectively; rest achieves only 0.45, driven by a recall of 0.36. The low rest recall is the primary driver of the overall performance ceiling.

3.4 Computational Performance

The following section will analyse more in depth the computational performance.

The algorithmic latency of the pipeline was segmented into three components following the framework: $T_{\text{total}} = t_{\text{window}} + t_{\text{preprocessing}} + t_{\text{inference}}$.

The analysis window is $t_{\text{window}} = 1500$ ms and represents the fixed waiting time set by the ERD feature extraction window of 0.5-1.5 s post-onset. During that period of time the system buffers incoming EEG data before any processing procedures take place. Preprocessing time is measured by executing the per-trial channel normalization function 1000 times and averaging the elapsed wall-clock time. That yields an estimate of

$t_{\text{preprocessing}} = 0.105$ ms. Inference time is benchmarked in a similar way over 1000 calls to the trained classifier's predict function, resulting in $t_{\text{inference}} = 1.203$ ms. CSP2 inference is not separately measured. Given that the total latency is already dominated by $t_{\text{window}} = 1500$ ms, its contribution is irrelevant to the analysis. The online processing loop is summarized in Table 3.

Table 3: Decomposition of the pipeline's algorithmic latency for online operation.

Step	Operation	Estimated time
1. Buffer data	Acquire 1500 ms of EEG into a sliding window	1500 ms
2. Preprocessing	Per-channel normalisation	0.105 ms
3. CSP1 inference	Spatial filtering + LDA (rest vs. movement)	1.203 ms
4. CSP2 inference	Spatial filtering + LDA (left vs. right)	< 0.01 ms
Total		≈ 1501.3 ms

The total latency is dominated by the acquisition window. The computational cost of all processing steps combined is negligible at a value of ≈ 1.3 ms. With a time equal to 1501.3 ms, the pipeline exceeds the 500 ms threshold that is typically cited for responsive closed-loop BCI systems [3]. Figure 22 summarises the latency decomposition confirming that the window is the bottleneck.

t_{window} (ERD): 1500.0 ms (bottleneck)
 t_{window} (MRCP): 500.0 ms (no bottleneck, -0.5 to +0.5 s)
 $t_{\text{preprocessing}}$: 0.105 ms
 $t_{\text{inference}}$: 1.203 ms
 Total latency: 1501.3 ms
 Real-time BCI suitable (< 500 ms): No - window dominates (Bottleneck)

Figure 22: Decomposition of the pipeline's algorithmic latency. The analysis window ($t_{\text{window}} = 1500$ ms) dominates entirely, while preprocessing (0.105 ms) and inference (1.203 ms) are negligible. The system is not suitable for real-time BCI operation in its current form.

In order to transform the latency into a real-time compatible range, a sliding-window approach could be adopted. Instead of waiting for a fixed 1500 ms epoch, the system classifies every 100-200 ms using a continuously updated buffer. For ERD features specifically, a moving mean of band-power computed over a short window of 500 ms would provide a continuously updated estimate of Mu and Beta de-synchronization, while eliminating the requirement for a fixed epoch. As a consequence the spectral estimation stability is reduced due to the shorter window, which would likely lower classification accuracy, however, it would also bring total system latency to the order of 100-200 ms. That would be a consistent result well within the range required for responsive closed-loop BCI operation.

4 Discussion

The objective of this study was to develop and evaluate a machine learning framework capable of classifying motor imagery EEG signals into three classes: left-hand imagery, right-hand imagery, and resting state. To achieve this goal, two ERD+MRCP-based models and several CSP-based configurations were compared under both within-subject and Leave-One-Subject-Out (LOSO) evaluation strategies.

The results demonstrate that CSP-based approaches consistently outperformed the handcrafted ERD+MRCP feature models. This finding agrees with previous motor imagery literature, where spatial filtering techniques have been shown to effectively enhance class separability by maximizing discriminative information across EEG channels. In particular, the best-performing configuration combined Mu and Beta band information with an eight-channel CSP representation, highlighting the importance of both spectral and spatial information for motor imagery decoding.

The comparison between Model 1 and Model 2 provides additional insight into the role of spatial information. While both models relied on the same ERD features, Model 2 incorporated MRCP features extracted from multiple electrodes instead of a reduced electrode set. Although the improvement in classification performance was relatively small, the results suggest that additional spatial information contributes positively to motor imagery discrimination. This observation supports the hypothesis that motor-related cortical activity is distributed across neighbouring sensorimotor regions rather than being confined to a single electrode location.

A substantial difference was observed between within-subject and LOSO evaluations. As expected, within-subject classification achieved considerably higher performance because the classifier was able to learn subject-specific neural patterns. In contrast, LOSO evaluation produced lower accuracies due to the large variability that exists between individuals. This result highlights one of the major challenges in EEG-based Brain-Computer Interfaces: developing models capable of generalizing across unseen users. Nevertheless, LOSO provides a more realistic estimate of real-world performance and should therefore be considered the most relevant evaluation framework for practical BCI deployment.

Analysis of the CSP cascade architecture revealed that the two classification stages exhibited different behaviours. The first stage, responsible for discriminat-

ing between Rest and Movement, consistently favoured LDA-based solutions, whereas the second stage, which separated Left and Right motor imagery, often benefited from alternative classifiers such as Logistic Regression. These findings suggest that different classification problems within the same pipeline may require different feature-classifier combinations and support the use of stage-specific optimization rather than a single classifier throughout the entire architecture.

The confusion matrix analysis revealed that the resting-state class represented the most challenging category. A substantial proportion of rest trials were misclassified as movement, indicating that the distinction between movement-related and non-movement activity remains a limiting factor of the current framework. This behaviour is consistent with the relatively weak discriminative power observed for some ERD and MRCP features and suggests that future work should focus on improving the detection of resting-state activity.

Several limitations should be acknowledged. First, EEG recordings are inherently noisy and exhibit considerable inter-subject variability, which directly impacts model robustness and generalization. Second, the dataset consisted of only 11 participants. Although such sample sizes are common in EEG research due to the complexity and duration of data acquisition, larger datasets would provide more reliable estimates of model performance. Furthermore, no dedicated validation set was used because allocating part of the dataset for validation would have significantly reduced the amount of data available for training and testing. This represents a common and generally accepted compromise in small EEG datasets, but it should nevertheless be considered when interpreting the reported results.

Finally, although the machine learning inference and preprocessing stages required only a few milliseconds, the overall latency of the system was dominated by the feature extraction window required for ERD computation. The current implementation introduces an approximate delay of 1.5 seconds, which would be excessive for practical real-time control. Consequently, future work should investigate sliding-window approaches and shorter analysis windows capable of reducing latency while maintaining classification performance.

Overall, the findings of this study confirm the effectiveness of CSP-based feature extraction for motor imagery classification and highlight the importance of spatial information, robust evaluation strategies, and latency-aware system design. These results contribute to the development of more reliable Brain-Computer Interface systems for neurorehabilitation and assistive

technology applications.

5 Conclusion

This study investigated the classification of motor imagery EEG signals using both handcrafted feature extraction approaches and CSP-based machine learning models. Two ERD+MRCP feature sets and multiple CSP configurations were evaluated under within-subject and Leave-One-Subject-Out (LOSO) validation frameworks in order to assess both subject-specific performance and cross-subject generalization.

The results demonstrated that CSP-based approaches consistently outperformed the ERD+MRCP models, confirming the importance of spatial filtering techniques for motor imagery decoding, as previously reported in the literature [Ramoser2000, 4, 5]. In particular, the best-performing configuration combined Mu and Beta band information with an eight-channel CSP representation, highlighting the complementary role of spectral and spatial information in EEG classification.

The study also revealed a substantial performance gap between within-subject and LOSO evaluations, emphasizing the strong inter-subject variability that characterizes motor imagery EEG signals. While within-subject models achieved higher accuracies by learning participant-specific neural patterns, LOSO evaluation provided a more realistic estimate of the performance expected in practical Brain-Computer Interface applications. This observation is consistent with previous studies addressing subject-independent EEG classification and BCI generalization challenges [6, 7, 8].

From a computational perspective, the proposed framework demonstrated very low preprocessing and inference times. However, the overall latency was dominated by the data acquisition window required for feature extraction, resulting in an approximate delay of 1.5 seconds. Although suitable for offline analysis, this latency remains excessive for real-time control and motivates the development of sliding-window approaches capable of significantly reducing response times. Such improvements are essential for future online Brain-Computer Interface systems [9, 10].

In summary, the findings of this work confirm the feasibility of decoding motor intentions from non-invasive EEG recordings and demonstrate the effectiveness of CSP-based feature extraction for multiclass motor imagery classification. The obtained results further support the relevance of ERD-based motor imagery paradigms originally described by Pfurtscheller and colleagues [2, 11]. Future work should focus on increas-

ing the number of participants, improving cross-subject generalization, reducing system latency, and validating the proposed framework in real-time neurorehabilitation and assistive technology applications. Additionally, deep learning approaches specifically designed for EEG analysis, such as EEGNet and related architectures, represent promising directions for future research [12, 13, 14, 15].

Use of Artificial Intelligence Tools

Artificial Intelligence (AI) tools, specifically Claude, were used during the preparation of this manuscript to assist with language refinement, code debugging, manuscript structuring, and scientific writing suggestions. All methodological decisions, experimental designs, analyses, results, figures, and interpretations were independently developed, verified, and validated by the authors. The authors take full responsibility for the content of this work.

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